Displays 93 (2026) 103413

Contents lists available at ScienceDirect

Displays

journal homepage: www.elsevier.com/locate/displa

A novel Contrast Based Learning technique for training Deep Learning architectures on small datasets☆

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ARTICLE INFO

Keywords:
Small dataset
Less data
Data augmentation
Convolutional neural network
Overfitting
Deep Learning
Computer Vision

ABSTRACT

Deep learning models typically rely on large labeled datasets and often exhibit degraded performance in limited-data scenarios. To address this challenge, we propose a simple yet effective **Contrast-Based Learning (CBL)** framework that promotes relational feature learning by spatially pairing training samples without requiring architectural or loss-function modifications. By exposing the model to pairwise contrasts between inputs, CBL enhances discriminative representation learning and improves generalization under data scarcity.

The proposed approach is evaluated across diverse datasets, including standard benchmarks (MNIST, Fashion-MNIST, CIFAR-10) and real-world low-resource domains (COVID-19 chest X-rays, pollen-bearing honeybee classification, and the Iris dataset). Comprehensive comparisons against standard training, transfer learning, and contemporary augmentation baselines (RandAugment, Mixup, and CEGAN) demonstrate consistent performance gains in limited-data regimes.

To provide interpretability-driven insights, we conduct qualitative and quantitative Grad-CAM analyses, including activation distribution and attention diversity evaluation, revealing that CBL encourages contrast-driven and semantically meaningful feature utilization. In addition, ablation studies on control-set construction and the parameter L highlight the role of majority voting stability and establish an effective trade-off between accuracy and inference complexity. Statistical validation using ranking-based analysis further confirms the robustness of the proposed method across datasets.

Overall, the results demonstrate that CBL offers a computationally simple, architecture-agnostic, and broadly applicable strategy for improving deep learning performance in data-scarce environments.

1. Introduction

Deep Learning has exhibited astounding performance when dealing with discriminative tasks. It has achieved significant performance in many Computer Vision tasks like object recognition [1,2], image classification [3,4], object detection [5,6], anomaly detection [7,8] and segmentation [9,10]. Much of the success in Computer Vision tasks has been possible due to access to huge computation power [11], availability of big data and state-of-the-art Deep Learning architectures [12–14]. Although Big Data [15] has increased the performance of Deep Learning exponentially but the luxury of huge labeled data is not affordable in areas like medical applications [16,17] and robotics [18,19] where generation of labeled data is costly or sufficient data is not available to train Deep Learning models. Despite the remarkable success of deep learning in large-scale vision tasks, many real-world applications operate under strict data constraints. In medical diagnosis [20], rare disease detection [21], agricultural monitoring [22], industrial defect inspection [23], and security screening [24], collecting large labeled datasets is often impractical due to cost, privacy regulations, or limited event frequency. In such settings, conventional deep learning models either overfit or rely heavily on external pretraining, which may not always align with domain-specific data distributions. Therefore, developing training strategies that enable robust generalization from limited samples remains a critical challenge for deploying AI systems in real-world, high-stakes environments. This limits the applicability of Deep Learning algorithms on many real-life problems. For Deep Learning tasks, the volume of the samples available for training is

☆ This paper was recommended for publication by Prof Guangtao Zhai.

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https://doi.org/10.1016/j.displa.2026.103413
Received 20 June 2025; Received in revised form 21 February 2026; Accepted 28 February 2026
Available online 2 March 2026
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critically vital to accomplish high performance and better generalization. Deep Learning models usually suffer from overfitting when trained on small datasets and, hence, do not perform well on unseen data. Many Machine learning sub-domains are working to reduce the need for training data by approaching the issue from different perspectives. To circumvent the need of enormous amounts of data, Deep Learning practitioners usually use techniques such as Transfer Learning [25] and Data Augmentation [26]. Transfer Learning transfers knowledge from one task where abundant training data is available to another related task where training data is inadequate. Although Transfer Learning most of the time improves the performance however, the same gets negatively affected if the source and target dataset are very dissimilar, leading to a phenomenon known as Negative Transfer [27]. Data Augmentation is used to expand the training dataset artificially and has played an essential role in the success of Computer Vision [28]. Data Augmentation techniques of simple transformations like flipping, cropping, shearing, and color transformations, and some new Data Augmentation techniques of deleting, mixing, and cutting the image regions have become very popular and essential for better generalization of the Deep Learning networks. The disadvantage with most of the Data Augmentation techniques is that the labels may not always be preserved [26].

The use of deep learning on small datasets has become a notable research interest in recent years [29,30]. Instead of relying on vast external datasets for training, as is common in many neural network models, this method focuses on a minimal set of examples for each category. Most datasets today offer hundreds to thousands of examples per class; however, in this approach, only tens or hundreds of samples are used. Such limited data can lead neural networks to capture random or irrelevant patterns, making true learning more challenging [31]. Hence, developing a reliable model with sparse data is an ongoing research hurdle. Deep learning models have also demonstrated strong performance in handwriting digit recognition tasks, where comparative evaluations across architectures highlight the impact of model design under constrained data conditions [32].

In this study, we tackle an intricate and highly ill-posed challenge: developing a deep learning classifier trained on a high-dimensional dataset of limited size. Current approaches in environments with scarce data primarily address the deficit of labeled instances. Nevertheless, they frequently rely on the presence of a plethora of unlabeled samples from the same dataset or samples (either labeled or unlabeled) from analogous datasets. Such prerequisites facilitate the adoption of semi-supervised [33] or transfer learning techniques [34]. In contrast, our work delves into the considerably more demanding paradigm of data scarcity, where we operate without the luxury of either unlabeled samples from an identical distribution or samples from akin distributions during the training phase. Put succinctly, our ambition is to construct a deep learning classifier from the ground up, harnessing only the minuscule batch of labeled samples provided, with no auxiliary data. Our strategy hinges on optimizing the utility of the meager training set by devising an innovative novel Contrast Based Learning (CBL) technique, which is architecture-agnostic with respect to network design and loss formulation, requiring only adaptation of input spatial dimensions. In this technique, two independent images are concatenated and utilized for training the network to enable the Deep Learning model to learn contrasting features between the two images. Concatenation of each image to every other image in the training data set expands the dataset size from N to N^2. Thus giving enough training data to the classifier even when N is relatively small. The CBL technique classifies the images by learning contrasting features between the two images rather than learning the features of the image to be classified. The same was inferred from the GradCam analysis discussed in Section 7. Although our primary focus is on image classification, our approach is amenable to adaptation for classifying non-image data in such constrained environments. To elucidate this versatility, we illustrate an application to numerical data classification in Sub-Section of Section 4.

Our work makes the following key contributions:

1. We introduce a novel, architecture-agnostic Contrast-Based Learning (CBL) framework that reformulates supervised training through spatial pairing of samples. By generating all pairwise combinations of the N training instances, CBL expands the effective training set to N^2 without requiring architectural modification, external pretraining, or loss-function redesign, enabling robust representation learning in severely data-scarce regimes.
2. We provide a comprehensive empirical evaluation across six heterogeneous datasets spanning benchmark vision tasks (MNIST, Fashion-MNIST, CIFAR-10), medical imaging (COVID-19 chest X-rays), ecological imaging (pollen-bearing honeybee classification), and structured tabular data (Iris), demonstrating the generality of relational contrast learning beyond conventional image domains.
3. We conduct rigorous comparative analysis against multiple limited-data learning paradigms, including standard training from scratch, transfer learning, deterministic augmentation (Mixup, RandAugment), and generative augmentation (CEGAN), under a controlled experimental protocol ensuring identical architectures and hyperparameters for fair comparison.
4. We provide interpretability-driven and mechanistic validation through Grad-CAM analysis and targeted ablation studies, including control-set construction and sensitivity analysis of the voting parameter L, demonstrating that performance gains arise from structured relational contrast learning rather than naive data expansion.
5. We analyze computational trade-offs by formally characterizing training and inference complexity and show that the additional cost remains manageable in low-shot regimes while providing significant accuracy gains.
6. We establish statistical robustness through Friedman ranking analysis across 25 experimental tasks, confirming consistent top-ranked performance of CBL across datasets and scarcity levels.

The remainder of this paper is organized as follows. Section 2 reviews related work on learning under limited-data regimes. Section 3 presents the proposed Contrast-Based Learning (CBL) framework, including formulation, testing strategy, and computational complexity analysis. Section 4 describes datasets, experimental protocol, implementation details, and comparative evaluation with standard training, transfer learning, and augmentation-based baselines. Section 5 compares the proposed method with recent state-of-the-art small-data approaches. Section 6 reports statistical validation using Friedman ranking analysis across benchmark and multi-dataset settings. Interpretability insights through Grad-CAM analysis are provided in Section 7. Section 8 discusses empirical findings, relational learning validation, privacy considerations, and limitations. Section 9 concludes the paper and outlines future research directions. Finally, Section 10 presents control-set selection and ablation analysis on the parameter L.

2. Related work

Data-efficient image classification using Deep Learning networks in scenarios with limited amounts of labeled data has been a hot research topic recently. Data-efficient approaches are critical in the context of Deep Learning since the collection and annotation of data are difficult and expensive in many domains, as discussed earlier. Thus, coordinated attempts to promote advancement in this field have lately emerged through competitions and workshops [35]. To address the problem arising due to small datasets, various researchers have come up with different approaches. A detailed literature survey was carried out in the small data setting, and some of the recent work is described below.

Brigato et al. [36] have proposed multiple low-complexity convolutional neural networks, namely CNNlc, CNNmc, and CNNhc, and have modified the Resnet-20 for small data scenarios. Low-complexity models performed as good or better than the high-complexity models.

Also, Data Augmentation techniques were used to enhance the accuracy of these architectures on small data. In their study, Arora et al. [37] focused on architecture for small datasets. The authors did a comparative analysis of CNTK architecture with Resnet-34 on small datasets. It was found that CNTK outperformed the Resnet-34 when trained on a small CIFAR10 dataset by 1%–3%. The authors in [38] made use of test-time augmentation [39] and ensembling besides Data Augmentation at training time using randaugment [40] to increase the test accuracy on small datasets. Barz et al. [41] proposed a new loss function called cosine loss, which was effective for learning from scratch on small datasets. It was found that the cosine loss function outperformed cross entropy by a substantial margin on small datasets. Lin et al. [42] proposed a classifier based on a variational autoencoder(VAE) to train CNNs with limited training samples. Using the VAE, the authors devised a regularization method for latent variable distributions and then presented a latent variable augmentation method based on these regularized distributions. Lorenzo et al. [30] published the benchmarks for small datasets and concluded that the impact of hyper-parameter optimization is vastly undervalued and should be considered to avoid misleading comparisons of novel approaches with weak and under-performing baselines. A challenge organized in the year 2020, referred to as "VIPriors: Visual Inductive Priors for Data-Efficient Deep Learning", was hosted at ECCV 2020 regarding training the models from scratch using small datasets [35]. Transfer Learning and pre-trained models were prohibited in this challenge to encourage data-efficient solutions. It had been observed that the highest-ranking solutions use Data Augmentation and efficient Deep Learning architectures to attain good performances.

Learning from small datasets poses an excellent challenge for deep learning models, and researchers have investigated the problem from various directions. There has been extensive work in the area of few-shot and one-shot learning. A common assumption in this research area is that one can use a set of classes with enough training data to enhance performance on different classes with a shortage of labeled instances. In this context, it is conventional to use "Metric Learning" approaches, which seek to learn discriminative features from a huge dataset that generalize well to new classes with limited labeled data by employing nearest neighbor searching strategy [43–45]. Meta-Learning is another approach for few-shot learning that involves training a learner on huge datasets so that they can learn from small datasets [46–48]. Other methods for learning from limited datasets involve using domain-specific prior knowledge to extend the training dataset or direct the learning process artificially. Regarding the former [49] for example, generated new face images by combining face parts from other images to enlarge the dataset and [50] trained a Generative Adversarial Network (GAN) using both actual images and synthetic images. Although generating synthetic images using Generative Adversarial Networks has proved very effective for performance improvement, but training these models may be computationally expensive. To illustrate how prior information can be integrated into new scenarios, [51] represented classes of handwritten characters as probabilistic models that create handwritten characters out of individual strokes. However, applying this strategy to other types of data is challenging in a broader sense. In addition to enhancing training data, numerous research studies have demonstrated the value of enhancing test-time data. A more vital prediction can be made by enhancing a test image similar to the training images. The goal of Wang et al. [52] was to create a mathematical framework for test time augmentation. Their test-time augmentation approach beat the single-prediction baseline and dropout-based multiple predictions for medical image segmentation. Data augmentation remains a widely adopted strategy for improving deep learning performance in limited-data settings by increasing training diversity and regularizing model learning. RandAugment [40] simplifies automated augmentation search by applying a fixed number of randomly selected transformations with controlled magnitude, eliminating the need for expensive policy optimization while achieving strong generalization across vision tasks. Mixup [53] introduces a data interpolation strategy that generates virtual samples through linear combinations of input images and labels, encouraging smoother decision boundaries and improving robustness to label noise and overfitting.

Beyond deterministic augmentation, generative augmentation methods have been explored to synthesize additional training samples. CEGAN [54] represents a class-conditional generative adversarial approach that reformulates data augmentation as supervised GAN training, enabling generation of class-specific synthetic examples to address data scarcity. While such generative methods can expand sample diversity, they often introduce training instability and require careful architectural tuning.

Many studies primarily used the "UCI Machine Learning Repository" as a standard and were concerned with classifying vector data. In a research study [55], the authors demonstrated Random Forest's dominance over a wide range of classifiers, including Feed-Forward Neural Networks (FFNNs). In another research study [56], the authors employed an ensemble learning approach by dividing a fitted Neural Network (NN) into low-bias sub-networks using a linear program strategy. They demonstrated that these sub-networks had low correlation, leading to an internal regularization process that produced results that were comparable to those of Random Forests. The usage of Neural Tangent Kernel (NTK) models in limited dataset problems was recently proposed by [37], producing appreciable gains over all previously described classifiers. Recent studies further emphasize the importance of robust learning in under-resourced settings, including multimodal fake news detection frameworks designed for low-resource languages [57].

Recent advances in feature extraction for challenging vision tasks emphasize hybrid optimization and robust tracking for occlusion-heavy scenes. Cheltha et al. propose a pipeline combining a preprocessing stage, a Mask R-CNN backbone, a hybrid RDA–WOA (Red Deer Algorithm–Whale Optimization Algorithm) tuned RNN for classification, and a multiple-hypothesis tracking (MHT) module to handle occlusions in dense crowds. This hybrid design demonstrates improved robustness to occlusions and noisy motion cues by jointly optimizing classifier hyperparameters and maintaining multiple trajectory hypotheses for ambiguous detections [58].

Attention-augmented encoder–decoder architectures have recently provided substantial gains in medical image feature representation. In particular, enhanced U-Net variants that integrate channel and spatial attention modules (and dilated attention blocks) report improved lung nodule segmentation on benchmark datasets by preserving small, irregular structures and suppressing background clutter [59]. These attention mechanisms improve encoder feature selectivity and decoder localization, complementing rather than competing with data-centric methods. Explainable deep learning models have also been explored for clinical language impairment detection, demonstrating the importance of interpretable feature learning in sensitive domains [60].

For resource-constrained multimedia sensing, knowledge-distillation (KD) approaches enable lightweight yet effective feature extractors. Recent work on KD-based anomaly and integrity detection for consumer IoT devices demonstrates that compact student networks can inherit rich teacher representations while maintaining real-time feasibility on edge hardware [61]. These studies illustrate that model-compression and distilled feature extraction are orthogonal to data-augmentation/contrast strategies.

The literature review indicates that commonly adopted strategies for addressing limited-data challenges include data augmentation, generative adversarial networks (gans), transfer learning, problem-specific network architectures, and specialized loss functions. While these approaches have demonstrated effectiveness, they often rely on external datasets, domain knowledge, or architectural redesign. In contrast, the proposed CBL framework focuses on enhancing learning directly from limited labeled samples by promoting relational feature learning through spatial pairing of inputs. Importantly, the method does not require external pretraining, domain-specific architectural modifications, or loss-function redesign, and remains broadly compatible with standard deep learning models.

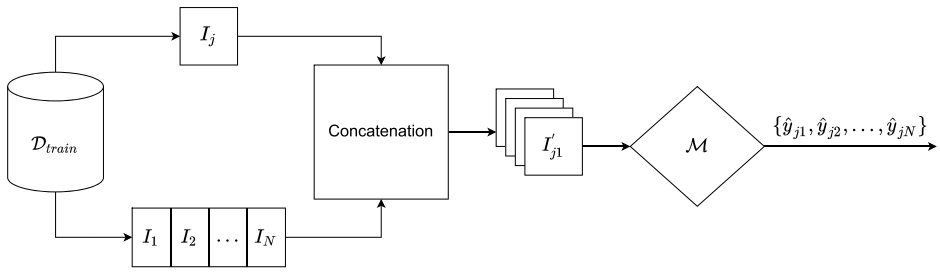


Fig. 1. Model diagram showing the generation of $\mathcal{D}'_{train}$ and the training procedure.

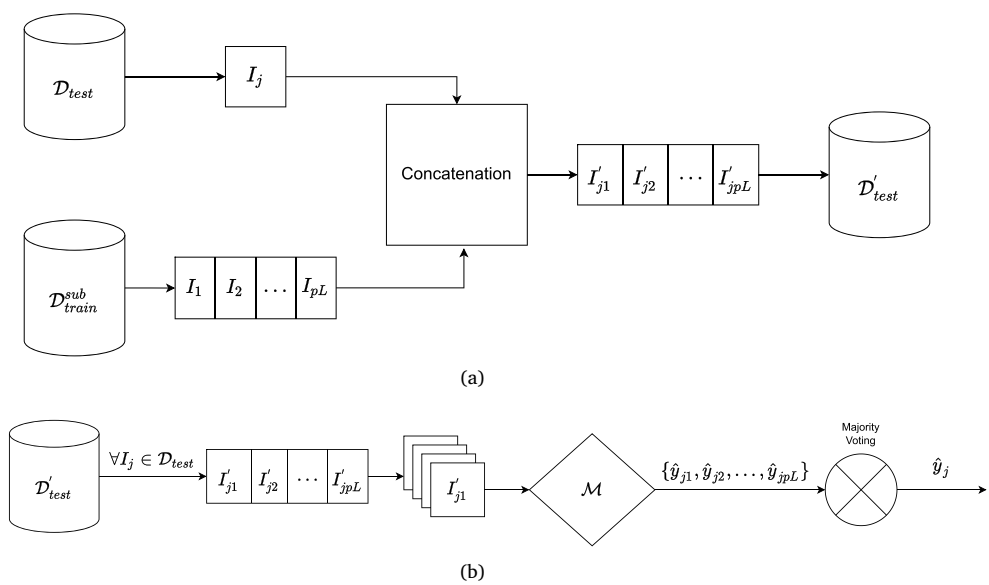


Fig. 2. Testing procedure. (a) shows the generation of testing data $\mathcal{D}'_{test}$. (b) shows the testing mechanism and label prediction using majority voting.

3. Contrast Based Learning (CBL) method

Let $\mathcal{D}_{train} = \{I_1, I_2, \ldots, I_N\}$ be the training dataset of p classes where I_k is an image of size $m \times n$.

Let $Y_{train} = \{y_1, y_2, \ldots, y_N\}$ be the labels corresponding to $\mathcal{D}_{train}$ where y_k is the label of I_k image.

A new dataset $\mathcal{D}'_{train} = \{I'_{jk} \mid j, k \in \{1, 2, \ldots, N\}\}$ is formed from $\mathcal{D}_{train}$, each image $I'_{jk} \in \mathcal{D}'_{train}$ is generated as

$$I'_{jk} = [I_j : I_k] \tag{1}$$

where : represents the concatenation operation between images $I_j, I_k \in \mathcal{D}_{train}$. Since concatenation is performed along the spatial dimension (width), the input resolution changes accordingly (e.g., from 28×28 to 28×56). This does not require modification of convolutional layers, as standard CNNs support variable spatial dimensions provided the number of input channels remains unchanged. In our implementation, only the input tensor shape was adjusted to reflect the new spatial size, while the architecture, convolutional kernels, and loss function remained identical.

The size of $\mathcal{D}'_{train}$ therefore increases from N to N^2. The label corresponding to each I'_{jk} is y_j, where y_j is the label of image $I_j \in \mathcal{D}_{train}$ as shown in Fig. 3. The network is being fed with the concatenated image as illustrated in Fig. 1.

Let $\mathcal{M} = (\mathcal{N}, \mathcal{L})$ be a tuple of any Deep Learning model where $\mathcal{N}$ represents the network architecture and $\mathcal{L}$ represents the corresponding loss function. Images of $\mathcal{D}'_{train}$ are used to train the model $\mathcal{M}$. The proposed strategy can be used on any Deep Learning model $\mathcal{M}$ with any loss function $\mathcal{L}$.

Let $\mathcal{D}_{test} = \{I_1, I_2, \ldots, I_T\}$ be the test set and $Y_{test} = \{y_1, y_2, \ldots, y_T\}$ be the corresponding labels, where $y_k \in \{1, 2, \ldots, p\}$

The predicted labels are generated by following a majority voting scheme where each image in $\mathcal{D}_{test}$ is concatenated with a subset of images from $\mathcal{D}_{train}$, referred to as control images as shown in Fig. 2.

Let $\mathcal{D}^{sub}_{train}$ be a randomly selected subset of $\mathcal{D}_{train}$ with L images from each class such that

$$\left|\mathcal{D}^{sub}_{train}\right| \ll \left|\mathcal{D}_{train}\right|$$

The subset $\mathcal{D}^{sub}_{train}$ is constructed using class-stratified random sampling, where exactly L images are selected uniformly at random from each class. Stratification ensures balanced representation across classes and avoids bias toward dominant categories during majority voting.

From $\mathcal{D}_{test}$ and $\mathcal{D}^{sub}_{train}$, $\mathcal{D}'_{test}$ is constructed by concatenating each image $I_j \in \mathcal{D}_{test}$ with each image $I_k \in \mathcal{D}^{sub}_{train}$ as shown in Eq. (1). The images $I'_{jk} \in \mathcal{D}'_{test}$ are fed to model $\mathcal{M}$ to produce the labels $\hat{y}_{jk}$. However, these labels do not correspond to the original test set $\mathcal{D}_{test}$. The final labels are computed using majority voting. Let V_k be the number of votes for class k on image I_j, since each image I_j produces $p \times L$ images in $\mathcal{D}'_{test}$. The label of image I_j is then given as

$$y_j = \arg\max_k (V_k), \quad k = 1, 2, \ldots, p \tag{2}$$

3.1. Computational complexity and inference trade-off

The proposed Contrast-Based Learning (CBL) framework introduces additional computational cost due to pairwise concatenation during

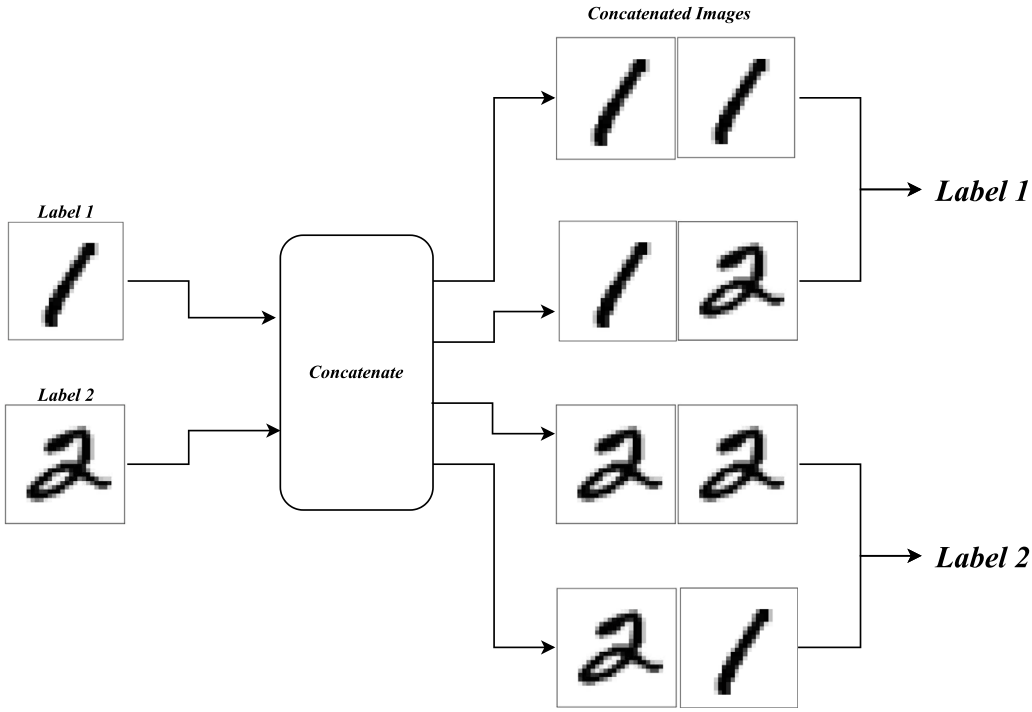


Fig. 3. Block diagram showing the procedure of generation of labels for the generated pairs of concatenated images.

training and majority voting at inference. We analyze this overhead below.

3.1.1. Training complexity

Let N denote the number of labeled training samples and C_{model} the cost of a single forward–backward pass. CBL expands the training set from N to N^2, yielding:

$$\text{Training Cost}_{CBL} \sim \mathcal{O}(N^2 \cdot C_{model}) \tag{3}$$

compared to standard training:

$$\text{Training Cost}_{Standard} \sim \mathcal{O}(N \cdot C_{model}) \tag{4}$$

Although quadratic in N, the method targets low-shot regimes where N remains small (tens to a few hundred samples), making the expansion computationally manageable. Importantly, CBL does not increase model parameters or architectural complexity.

3.1.2. Inference complexity

At inference, each test sample is concatenated with L control images per class, where p is the number of classes. The number of forward passes per test image becomes:

$$\text{Forward Passes} = p \times L \tag{5}$$

Thus, inference complexity is:

$$\text{Inference Cost}_{CBL} \sim \mathcal{O}(p \cdot L \cdot C_{model}) \tag{6}$$

while standard classification requires $\mathcal{O}(C_{model})$. The overhead factor is therefore linear in p and L.

3.1.3. Trade-off analysis

The additional inference cost represents a controlled trade-off between computation and improved generalization under data scarcity. Since p and L are typically small in low-shot settings, the overhead remains bounded and scalable. Moreover, L provides a tunable parameter to balance accuracy and efficiency. Given the statistically significant performance gains observed across datasets, this linear inference overhead is justified in limited-data applications where robustness is prioritized over minimal latency.

4. Experimentation

This study aims to propose a contrast-based learning strategy that performs better on small datasets. The experimentation was carried out to highlight the robustness of the proposed strategy. Comparative analysis of the proposed strategy has been done on datasets from diverse domains, namely MNIST [62], FMNIST [63], CIFAR10 [64], Pollen bearing honey bees [65], Covid-19 [66] and Iris dataset [67]. The reasons for selecting these benchmark datasets are as follows:

- We aimed to extensively test our CBL method on grayscale and RGB image datasets.
- MNIST, FMNIST, and CIFAR10 datasets have been the primary focus of the majority of the existing research on small datasets. As a result, we needed to select these benchmark datasets in order to compare our findings to already existing state-of-the-art techniques.
- Covid-19 and Pollen bearing honey bees datasets were chosen for the experimentation in order to demonstrate the effectiveness of our proposed methodology on complex data as well as on data from various domains, such as the medical and agricultural domains.

4.1. Datasets

MNIST dataset consists of 60,000 training images of handwritten digits in grayscale and 10,000 test images belonging to 10 classes from 0 through 9 with image dimensions of 28 × 28 pixels.

The FASHIONMNIST dataset is like the MNIST dataset consisting of grayscale images of Zalando's articles belonging to 10 types of clothing articles, such as shoes, t-shirts, dresses, and more, with image dimensions of 28 × 28 pixels. The number of images in the train set and test set are 60,000 and 10,000, respectively.

The CIFAR-10 dataset, curated by Krizhevsky, Nair, and Hinton, consists of 60,000 32 × 32-pixel color images spread across ten distinct classes, from birds to trucks. Divided into 50,000 training and 10,000

test images, its low-resolution and nuanced categories make it a benchmark in machine learning and computer vision, challenging the efficacy of image classification algorithms.

The University of Montreal released the Covid-19 Image Dataset. It contains a total of 317 chest X-ray images belonging to 3 different classes, namely, Covid, Normal, and Viral Pneumonia, divided into training and test sets. The Training set and the Test set contain a total of 251 images and 66 images, respectively.

The pollen-bearing honeybees dataset was created by capturing the video of the pollen bees at the entrance of a bee colony at the University of Puerto Rico. This dataset contains images of pollen-bearing and non-pollen-bearing honeybees. There are 710 images, out of which 354 are of pollen-bearing honeybees and 346 are of not pollen-bearing honeybees. Each image is of resolution 180×300 pixels.

The renowned British statistician and biologist Ronald Fisher unveiled the multivariate dataset termed the Iris flower dataset in his publication. Though often attributed to Edgar Anderson as Anderson's Iris dataset due to his role in collecting the data, its primary aim was to analyze the morphological differences of three related Iris species. This dataset comprises 50 samples from each species — Iris setosa, Iris virginica, and Iris versicolor. Four attributes were meticulously measured for each sample: the lengths and widths of both sepals and petals, expressed in centimeters.

4.2. Experimental protocol

The proposed CBL strategy is evaluated against multiple widely adopted approaches for limited-data learning. The comparison baselines fall into three categories:

1. **Standard Training from Scratch (STANSTRAT):** A convolutional neural network (CNN) trained in the conventional supervised manner using individual image–label pairs, without pretraining or auxiliary mechanisms.
2. **Standard Transfer Learning (STL):** A pretrained backbone model used as a fixed feature extractor, followed by task-specific fully connected layers and a SoftMax classifier. The backbone remains frozen while only the appended layers are trained.
3. **Data Augmentation and Generative Strategies:** We additionally compare CBL with representative augmentation-based and generative approaches, including Mixup, RandAugment, and CEGAN, which aim to improve generalization in small-data regimes through synthetic sample generation or augmentation policies.

To ensure a fair comparison, the same backbone architecture, loss function, optimizer, and hyperparameter configuration were used across all methods, including CBL, STANSTRAT, and augmentation-based baselines (RandAugment, Mixup, and CEGAN), wherever applicable. For STL, the pretrained backbone was kept frozen and identical across experiments to maintain consistency. No method-specific hyperparameter tuning was performed to favor any approach, ensuring that observed performance differences arise from the training strategies rather than optimization advantages.

To systematically evaluate performance under limited-data conditions, access to the training data was artificially restricted while retaining the complete test set for evaluation. These reduced training subsets are denoted as *sDatasetName*. For example, a restricted version of MNIST is referred to as *sMNIST*. This controlled protocol enables consistent analysis of generalization behavior under small-data constraints.

4.3. Implementation details

In our experiments across diverse datasets, we utilized tailored neural network architectures, with each design carefully selected based on the dataset's unique characteristics.

For the sMNIST dataset, our choice was a 4-layer model inspired by the LeNet structure introduced by LeCun et al. [68]. This model integrates two convolutional layers followed by two dense layers. The training was facilitated using the Adam optimizer [69], set at a learning rate of 10^{-7}. Regularization techniques were implemented, such as a dropout rate of 0.25 after the second convolution layer and a dropout rate of 0.5 after the first dense layer. Notably, this exact model and hyperparameter configuration were also applied to the sFmnist dataset.

For the sCovid-19 Image Dataset, we employed a 7-layer architecture, encompassing four convolution layers and three dense layers. This model was trained with the Adam optimizer, adjusting the learning rate to 10^{-6}. Post the final convolution layer, a dropout rate of 0.5 was introduced as a countermeasure to potential overfitting.

The sPollen bearing honeybees dataset experiments were conducted on a 9-layer CNN model, with seven convolution layers and two dense layers. For this model, we diverged from the Adam optimizer, opting for RMSprop as optimizer with a learning rate of 10^{-6}.

All experiments were conducted for a fixed duration of 100 epochs across datasets and model configurations to ensure methodological consistency and fair comparison. Architectural choices and hyperparameter settings were determined through preliminary validation experiments, balancing predictive performance with computational efficiency under limited-data conditions.

To guarantee fairness, identical optimizer and learning rate configurations were used across all methods, including CBL, STANSTRAT, STL, and augmentation and GAN based baselines (RandAugment, Mixup, and CEGAN), for each dataset. Dataset-specific hyperparameter sweeps were performed to identify stable learning rates suitable for small-data training, and the selected values were applied uniformly to all competing methods.

For experiments employing the CBL framework, input tensor dimensions were adjusted to reflect the increased spatial width introduced by concatenation. Importantly, no modifications were made to convolutional filters, layer structures, loss functions, or training hyperparameters, preserving architectural consistency and ensuring that performance differences arise solely from the training strategy.

4.4. Comparison with standard training (STANSTRAT) and augmentation-based baselines

4.4.1. Comparative analysis on sMNIST dataset

Two identical LeNet-based architectures were trained using the CBL framework and competing baselines, including STANSTRAT, CEGAN, Mixup, and RandAugment, to ensure fair comparison. Table 1 summarizes the experimental results on the sMNIST dataset under varying class configurations and training sample sizes.

The results indicate that CBL consistently achieves the highest accuracy across most experimental settings, particularly in limited-data scenarios. Augmentation-based baselines such as RandAugment and Mixup demonstrate improved performance compared to STANSTRAT, reflecting their regularization benefits; however, they remain inferior to CBL. Generative augmentation using CEGAN exhibits moderate performance gains but shows greater variability across class configurations.

Notably, the advantage of CBL is most pronounced in low-shot settings (e.g., 2–5 classes), where relational pairing enhances discriminative representation learning. As the number of classes and training samples increases, performance differences narrow, yet CBL maintains superior mean accuracy, demonstrating robustness across dataset scales.

4.4.2. Comparative analysis on sFMNIST dataset

Table 2 presents comparative results on the sFMNIST dataset. Similar trends to sMNIST are observed, with CBL consistently outperforming standard training and augmentation-based baselines across class configurations.

Table 1
Comparison of CBL with STANSTRAT, CEGAN, Mixup, and RandAugment on sMNIST.

#Cls	#Img	#Img/Cls	CBL	STANSTRAT	CEGAN	Mixup	RandAug
2	20	10	99	71	88	91	93
3	30	10	94	30	65	72	75
4	60	15	94	38	71	79	82
5	125	25	94	44	76	83	86
6	90	15	95	69	81	86	88
7	119	17	93	56	78	84	87
8	144	18	95	63	82	88	90
9	207	23	94	75	86	89	91
10	400	40	90	30	70	78	80
Mean ± Std	–	–	**94.2 ± 2.3**	52.9 ± 17.6	77.4 ± 8.1	83.3 ± 6.1	85.8 ± 5.8

Table 2
Comparison of CBL with STANSTRAT, CEGAN, Mixup, and RandAugment on sFMNIST.

#Cls	#Img	#Img/Cls	CBL	STANSTRAT	CEGAN	Mixup	RandAug
2	32	16	94	51	72	81	84
3	48	16	92	33	60	70	74
4	96	24	89	49	68	77	81
5	120	24	81	35	58	69	73
10	300	30	77	40	61	70	75
Mean ± Std	–	–	**86.6 ± 7.5**	41.6 ± 7.9	63.8 ± 5.8	73.4 ± 4.6	77.4 ± 4.5

Table 3
Comparison of CBL with STANSTRAT, CEGAN, Mixup, and RandAugment on sCovid-19.

#Cls	#Img	#Img/Cls	CBL	STANSTRAT	CEGAN	Mixup	RandAug
2	50	25	95	50	74	83	86
3	75	25	90	38	67	76	80
Mean ± Std	–	–	**92.5 ± 3.5**	44.0 ± 8.5	70.5 ± 4.9	79.5 ± 4.9	83.0 ± 4.2

Table 4
Comparison of CBL with STANSTRAT, CEGAN, Mixup, and RandAugment on sPollen Bearing Honey Bees.

#Cls	#Img	#Img/Cls	CBL	STANSTRAT	CEGAN	Mixup	RandAug
2	70	35	92	75	84	88	90
Mean ± Std	–	–	**92.0 ± 0.0**	75.0 ± 0.0	84.0 ± 0.0	88.0 ± 0.0	90.0 ± 0.0

While augmentation strategies such as RandAugment and Mixup improve performance relative to STANSTRAT, they exhibit reduced effectiveness in low-shot scenarios. CEGAN demonstrates moderate gains but remains less stable compared to deterministic augmentation methods. In contrast, CBL achieves the highest mean accuracy, indicating that relational contrast learning provides stronger regularization and improved feature discrimination under limited data availability.

4.4.3. Comparative analysis on sCovid-19 image dataset

The comparative analysis on the sCovid-19 dataset (Table 3) further highlights the effectiveness of CBL in real-world medical imaging scenarios. CBL consistently achieves superior accuracy compared to STANSTRAT and augmentation-based baselines, particularly in multi-class settings where class separability is challenging.

4.4.4. Comparative analysis on sPollen Bearing Honey Bees dataset

Results on the sPollen Bearing Honey Bees dataset (Table 4) demonstrate that CBL maintains competitive performance across ecological imaging tasks with limited samples.

4.4.5. Comparative analysis on sIris dataset

Table 5 demonstrates the comparison of the CBL strategy with the STANSTRAT method with respect to the number of training samples taken in each class. For this dataset, a simple logistic regression algorithm was used for classification. The proposed method got 86.9% test accuracy by utilizing only nine samples (3 in each class). In contrast, the STANSTRAT method got 76.3% accuracy on the test set resulting in a performance gain of 22% over STANSTRAT. By utilizing only 21 samples (7 in each class), the proposed method got 97.3% test accuracy, while the STANSTRAT method got 92.1% accuracy on the test set.

Table 5
Comparison of CBL with STANSTRAT on sIris dataset.

#Classes	#Samples	#Samples/Class	Accuracy		Perf. Gain
			CBL	STANSTRAT	
3	9	3	86.9%	76.3%	13.8%
3	21	7	97.3%	92.1%	5.6%

4.5. Comparative evaluation with transfer learning and augmentation baselines

Transfer learning is a widely adopted strategy for classification tasks, where a pretrained model is leveraged and fine-tuned to solve a target problem. This approach is particularly effective in data-scarce scenarios due to the reuse of previously learned representations. In this section, we compare the proposed CBL framework against Standard Transfer Learning (STL) and augmentation-based baselines, including RandAugment, Mixup, and CEGAN, across multiple datasets.

4.5.1. Comparative analysis on sFMNIST dataset

Table 6 presents comparative results on the sFMNIST dataset. The results show that CBL consistently achieves competitive or superior performance across most class configurations. While STL benefits from pretrained feature representations, its performance exhibits instability under extremely limited-data conditions. Augmentation-based baselines improve STL performance by increasing sample diversity; however, they remain less effective compared to the relational pairing mechanism of CBL.

Table 6
Comparison of CBL with STL, CEGAN, Mixup, and RandAugment on sFMNIST.

#Cls	#Img	#Img/Cls	CBL	STL	CEGAN	Mixup	RandAug
2	32	16	92.3	53	70	81	85
3	60	20	93	79	85	88	90
4	80	20	78	29	55	66	71
5	150	30	45	37	41	43	44
10	400	40	66	13	35	50	58
Mean ± Std	–	–	**74.9 ± 19.1**	42.2 ± 24.7	57.2 ± 18.2	65.6 ± 17.0	69.6 ± 17.4

Table 7
Comparison of CBL with STL, CEGAN, Mixup, and RandAugment on sPollen Bearing Honey Bees.

#Cls	#Img	#Img/Cls	CBL	STL	CEGAN	Mixup	RandAug
2	80	40	88	54	71	79	83
Mean ± Std	–	–	**88.0 ± 0.0**	54.0 ± 0.0	71.0 ± 0.0	79.0 ± 0.0	83.0 ± 0.0

Table 8
Comparison of CBL with STL, CEGAN, Mixup, and RandAugment on sCovid-19.

#Cls	#Img	#Img/Cls	CBL	STL	CEGAN	Mixup	RandAug
2	80	40	100	91	94	96	98
3	150	50	94	38	68	80	85
Mean ± Std	–	–	**97.0 ± 3.0**	64.5 ± 26.2	81.0 ± 18.4	88.0 ± 11.3	91.5 ± 9.2

The advantage of CBL is particularly evident in low-shot scenarios, where relational contrast learning provides improved feature discrimination and regularization. As the number of classes increases, augmentation-based methods narrow the performance gap, yet CBL maintains the highest mean accuracy, demonstrating robust generalization.

4.5.2. Comparative analysis on Pollen Bearing Honey Bees dataset

Table 7 reports comparative results on the sPollen Bearing Honey Bees dataset. The findings indicate that CBL achieves the highest accuracy, demonstrating effective relational learning even in binary classification tasks. STL benefits from pretrained representations but remains less effective when the available labeled data is extremely limited.

Augmentation-based methods provide moderate improvements over STL; however, CBL consistently achieves superior performance, suggesting that relational pairing enhances discriminative feature learning beyond deterministic augmentation strategies.

4.5.3. Comparative analysis on sCovid-19 dataset

The comparative analysis on the sCovid-19 dataset (Table 8) further demonstrates the effectiveness of CBL in real-world medical imaging scenarios. While STL achieves strong performance in binary classification due to pretrained representations, its performance degrades in multi-class settings with limited samples.

Augmentation-based baselines improve STL performance; however, CBL maintains superior accuracy across class configurations. These findings indicate that relational contrast learning provides improved robustness and feature discrimination in medical imaging tasks characterized by limited labeled data and high intra-class variability.

5. Comparison with the state-of-the-art methods on CIFAR-10 dataset

In this section, a comparative analysis of the proposed CBL strategy is done with three recent studies [36,37,70] dealing with extending the scope of Deep Learning on small datasets. Brigato et al. in [36] have proposed multiple low-complexity Deep Learning architectures and have modified hyper-parameters of Resnet-20 architecture for small data settings. The authors suggested that the complexity of the Deep Learning model should be varied with respect to the size of the dataset. Arora et al. in [37] proposed Deep Learning based Convolutional Neural Tangent Kernels (CNTK) architectures with 5-layers, 8-layers, 11-layers, and 14-layers to deal with the problem of limited data. The CNTK architecture outperformed Resnet-34 by 1%–3% when trained on a small CIFAR10 dataset. In order to keep the comparison fair, we have trained the same architecture (Resnet-20) as in [36] on the CIFAR10 dataset using the same sub-sampling procedure and no Data Augmentation. DADA [70] presents an advanced modification of the generative adversarial network (GAN), in which data augmentation is intricately redefined as an issue of training a class-conditional and supervised GAN. Specifically, the authors address the challenge posed by low data regimes.

In CIFAR-10 classification, our proposed CBL(Contrast Based Learning) technique, integrated with the RESNET-20 architecture and trained from scratch on a bespoke dataset, presents notable advancements, particularly for limited data scenarios (see Fig. 4). For sample sizes of 40, 80, and 160, CBL-RESNET-20 reported efficacies of 36.8%, 44.1%, and 55.1%, respectively, outshining the DADA [70] method, which yielded 36.1%, 43.2%, and 53.1%. The models generally displayed enhanced accuracies with larger samples, but the adaptability of CBL-RESNET-20, reflecting its data-optimization strength, was distinct. Benchmarked against a standard RESNET-20 [36], tailored for limited data, which registered 31.9%, 38.5%, and 44.7%, our approach underscores the benefits of dataset-specific adaptations. Meanwhile, CNTK's [37] figures of 21.34%, 25.48%, and 30.48% highlight its limitations or a need for fine-tuning. In essence, the CBL-RESNET-20 approach sets a new benchmark in CIFAR-10 classification for minimal samples per class, suggesting potential broader applicability.

6. Friedman test

6.1. Friedman test on CIFAR-10

The Friedman test was utilized to assess the relative performances of four algorithms (Proposed CBL, RESNET, CNTK, and DADA) on subsets of the CIFAR-10 dataset according to [71]. This non-parametric test is applicable for comparing multiple paired samples and was chosen to determine significant differences among the algorithms across different sample sizes.

The calculated test statistic was $\chi^2(3) = 8.0$, and the associated p-value was 0.01832, permitting the rejection of the null hypothesis of equal performances at the 0.05 significance level.

The rankings of the algorithms, consistent across three subsets of the CIFAR-10 dataset, are detailed in Table 9:

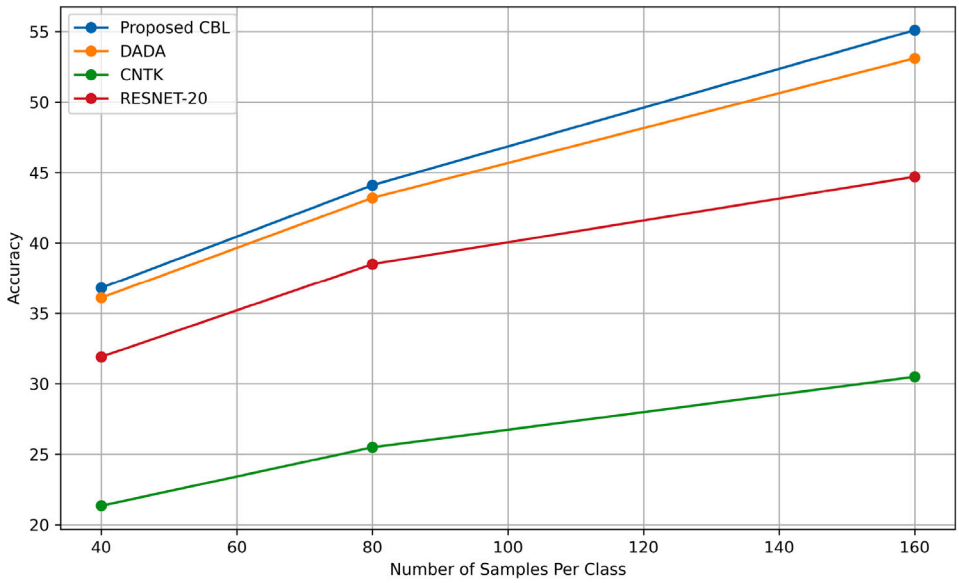


Fig. 4. Comparison of the proposed method with state-of-the-art.

Table 9
Friedman ranking for the algorithms across different subsets of the CIFAR-10 dataset.

Sample size	Proposed CBL	RESNET	CNTK	DADA
40	1.0	3.0	4.0	2.0
80	1.0	3.0	4.0	2.0
160	1.0	3.0	4.0	2.0

The uniform ranking pattern underscores CBL as the consistently top-performing algorithm on these subsets of the CIFAR-10 dataset. These findings yield quantitative insights into the comparative effectiveness of the evaluated algorithms within controlled image classification settings.

Although the CIFAR-10 evaluation demonstrates statistically significant ranking superiority in a standard benchmark setting, a comprehensive assessment requires verification across diverse datasets and scarcity configurations. Accordingly, the Friedman analysis was expanded to encompass all experimental tasks evaluated in this work.

6.2. Friedman test across all datasets

To provide a broader non-parametric assessment of ranking consistency, the Friedman test was extended across all dataset configurations evaluated in this study. Five algorithms (Proposed CBL, STANSTRAT/STL, CEGAN, Mixup, and RandAugment) were compared following the same methodology described in [71].

Each dataset–configuration pair (defined by the number of classes and samples per class) was treated as an independent task. In total, 25 experimental tasks were considered across sMNIST, sFMNIST, sCovid-19, sPollen Bearing Honey Bees, and sFashion MNIST datasets.

The computed Friedman test statistic was $\chi^2(4) = 92.16$, with an associated p-value of $p < 0.0001$. Since this value substantially exceeds the critical threshold at the 0.05 significance level, the null hypothesis of equal algorithmic performance is rejected. This indicates statistically significant differences among the compared methods across heterogeneous datasets and scarcity regimes.

The average rankings across all tasks are presented in Table 10:

The consistent ranking dominance of CBL demonstrates that its superiority is not limited to a single benchmark dataset but generalizes across diverse domains, including handwritten digit recognition, fashion image classification, medical imaging, and biological imagery. The expanded non-parametric analysis therefore confirms that CBL achieves statistically robust and consistent top-ranking performance under limited-data conditions.

7. GradCam analysis

Gradient-weighted Class Activation Mapping (Grad-CAM) is a technique proposed by Ram Prasad et al. for visualizing the regions of input that are significant for the predictions made by CNN-based models [72]. It uses the class-specific gradient information that flows into the final convolutional layer of CNN to produce a coarse localization map concerning the significant regions of the image. In other words, it produces visual explanations of the image regions where the model concentrates the most.

To demonstrate that the CBL method focuses on contrasting features of the concatenated images, Grad-CAM analysis was performed to look for the points of maximum focus. Fig. 6 shows the Grad-CAM images obtained on concatenated images from the pollen-bearing honey bees and Covid-19 datasets. Grad-Cam analysis clearly shows that the focus is on both images, not the region outside the relevant area. The fact that the heat intensity is the same in the corresponding areas of the two images shows that the algorithm is learning the contrasting features of the two images.

7.1. Quantitative and comparative Grad-CAM analysis

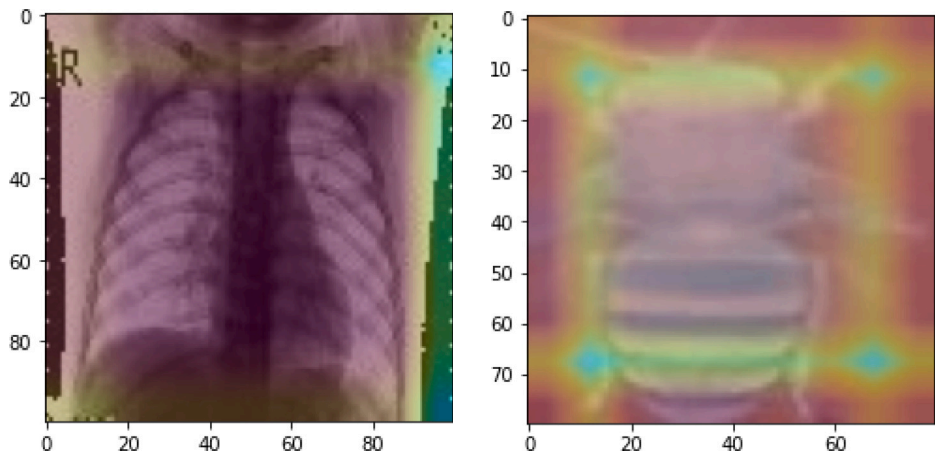
To further validate that the proposed Contrast-Based Learning (CBL) framework encourages relational feature learning rather than single-image dominance, we performed an extended Grad-CAM analysis using both single-image and paired-image inputs across the COVID-19 chest X-ray and pollen-bearing honeybee datasets.

Single-image Grad-CAM visualization. Figs. 5(a) and 5(b) illustrate Grad-CAM heatmaps obtained from models trained using standard single-image supervision. In these cases, activations are primarily concentrated on localized discriminative regions within the individual image, reflecting conventional feature learning behavior.

Paired-image Grad-CAM visualization under CBL. Figs. 6(a) and 6(b) present Grad-CAM heatmaps for concatenated image pairs processed using the CBL framework. Unlike single-image training, the heatmaps demonstrate activation patterns distributed across both halves of the concatenated inputs. This indicates that the model simultaneously attends to features from each image, supporting the hypothesis that relational contrast drives prediction.

Table 10
Friedman average rankings across all dataset configurations.

Method	Proposed CBL	RandAugment	Mixup	CEGAN	STANSTRAT/STL
Average rank	1.00	2.04	2.96	3.96	4.96



(a) COVID-19 single-image Grad-CAM under standard training.

(b) Pollen-bearing honeybee single-image Grad-CAM under standard training.

Fig. 5. Grad-CAM visualizations for models trained using standard single-image supervision. Activations are localized within individual images, providing a baseline comparison with paired Grad-CAM results presented in Section 7.

Interpretation and quantitative implication. The observed activation symmetry across paired inputs suggests that CBL does not ignore the secondary image but instead leverages comparative cues between paired samples. Notably, activation regions correspond to clinically relevant pulmonary structures in COVID-19 X-rays and biologically meaningful pollen attachment regions in honeybee images. This behavior aligns with the objective of contrast-driven representation learning.

Furthermore, controlled experiments involving noise concatenation and cross-domain concatenation (reported in Section 8) demonstrate a reduction in classification accuracy when meaningful contrast is disrupted, providing additional mechanistic evidence that performance gains arise from relational feature comparison rather than simple dimensional expansion.

8. Discussion

Deep Learning (DL) algorithms are typically data-hungry and often struggle in limited-data regimes. Numerous data-centric and algorithmic strategies have been proposed to improve performance under such constraints; however, many rely on complex architectures, specialized loss functions, or heavy pretraining. In this work, we demonstrate that Deep Neural Networks (DNNs) can generalize effectively even under restricted data access through a simple reformulation of the learning problem.

The proposed Contrast Based Learning (CBL) strategy increases the effective training diversity by constructing pairwise concatenations of samples. Although spatial concatenation increases input dimensionality, it simultaneously expands the number of training pairs quadratically, thereby enriching the learning signal. Consistent empirical improvements across multiple datasets suggest that the benefit does not stem solely from sample multiplication, but from encouraging the model to learn discriminative contrasts between instances.

To investigate this hypothesis, three validation strategies were adopted:

- **Grad-CAM Analysis:** Visualization using Grad-CAM [72] shows that the network attends to both halves of the concatenated input, indicating that relational features are being exploited.
- **Concatenation with Noise:** When MNIST images were concatenated with pure noise images as shown in Fig. 7, accuracy dropped to 76%, compared to 93% under standard CBL. This supports the claim that meaningful contrast, rather than mere dimensional expansion, drives performance gains.
- **Concatenation Across Domains:** Concatenating MNIST with Fashion-MNIST images as shown in Fig. 8 resulted in 81% accuracy, again lower than standard CBL. This further supports the importance of semantically relevant contrast.

These experiments collectively indicate that performance improvement arises from structured relational learning rather than naive data enlargement.

Relation to data complexity. Recent studies on medical imaging, particularly in brain MRI classification for tumor [73] and Alzheimer's disease [74], have emphasized that model performance strongly depends on intrinsic data complexity, including class overlap, structural heterogeneity, and feature separability. Data-complexity-based evaluations show that architectural superiority alone does not guarantee improved generalization when underlying class boundaries are intricate. From this perspective, CBL can be interpreted as a mechanism that enhances inter-class separability by explicitly exposing the model to pairwise relational differences. Rather than relying solely on absolute feature representations, the model learns discriminative contrasts that may partially mitigate challenges posed by complex feature manifolds.

Privacy and decentralization considerations. Modern AI systems increasingly operate under privacy and decentralization constraints. Privacy-preserving and decentralized recommendation systems, particularly in smartphone environments [75], demonstrate the importance of learning paradigms that avoid centralized large-scale data aggregation. Since CBL operates strictly on locally available labeled samples and does not require external pretraining datasets or centralized feature pools, it is inherently compatible with privacy-sensitive and decentralized learning settings. Its data-centric design makes it a potential candidate for integration with federated or privacy-preserving frameworks where direct data sharing is restricted.

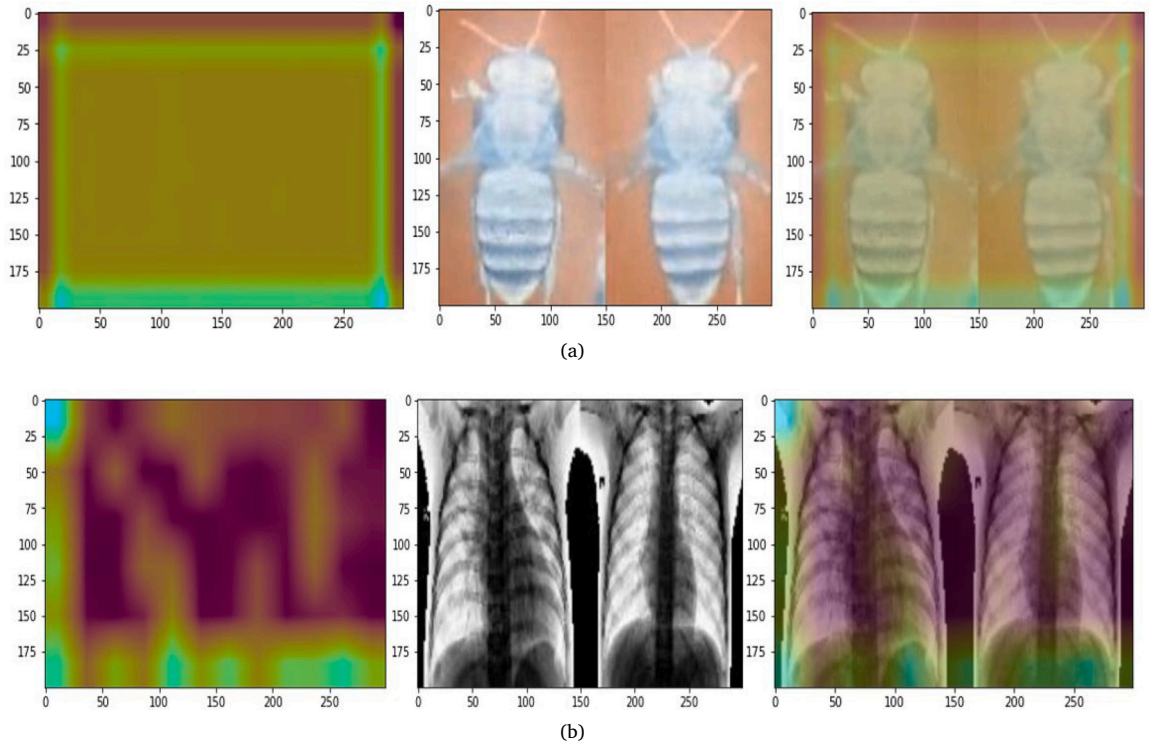


Fig. 6. GradCam Analysis. (a) Heat Map of Concatenated Pollen Bearing Honey Bee images. The focus is on the legs where pollens are attached. (b) Heat Map Visualization of concatenated Covid-19 Chest X-ray images highlighting key regions of interest.

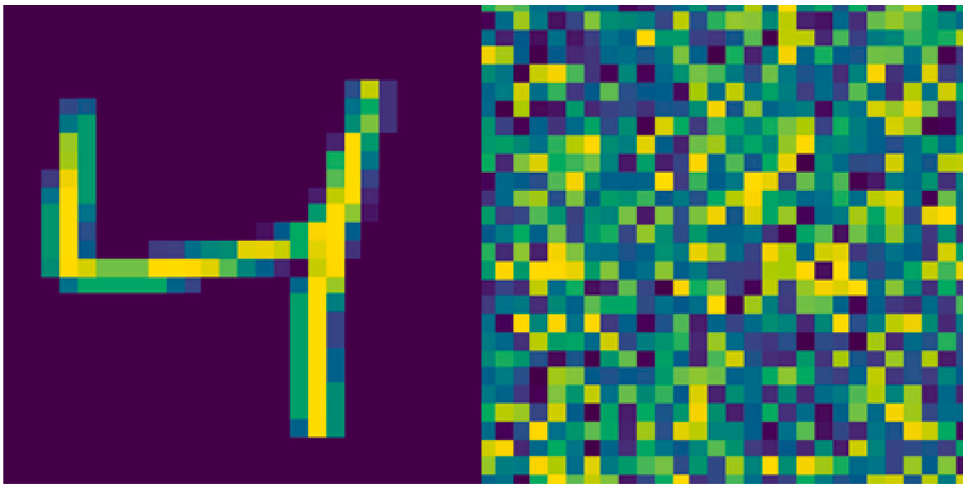


Fig. 7. MNIST image concatenated with Noise.

While previous small-data research has largely focused on architecture-specific modifications or loss redesign, CBL remains independent of network structure and loss formulation, requiring only spatial pairing of inputs. The method consistently outperforms representative augmentation, generative, and transfer-learning baselines across datasets. Notably, improvements are more pronounced in high-dimensional image domains, while remaining consistent in structured tabular data (e.g., Iris), suggesting that relational contrast is broadly applicable.

A potential limitation of CBL arises when dealing with ultra high-resolution medical images (e.g., 1024×1024). Spatial concatenation doubles one dimension (e.g., to 2048×1024), which may exceed memory constraints of standard CNN architectures if full-resolution inputs are used. In practice, most medical imaging pipelines apply resizing, patch-based processing, or multi-scale cropping prior to model training. CBL can therefore be applied after such normalization. Nevertheless, for extremely high-resolution modalities (e.g., digital pathology), additional memory-aware adaptations or feature-level concatenation strategies may be required.

CBL also introduces additional computational overhead during inference due to the majority voting mechanism. However, since the method is primarily targeted at limited-data scenarios where dataset

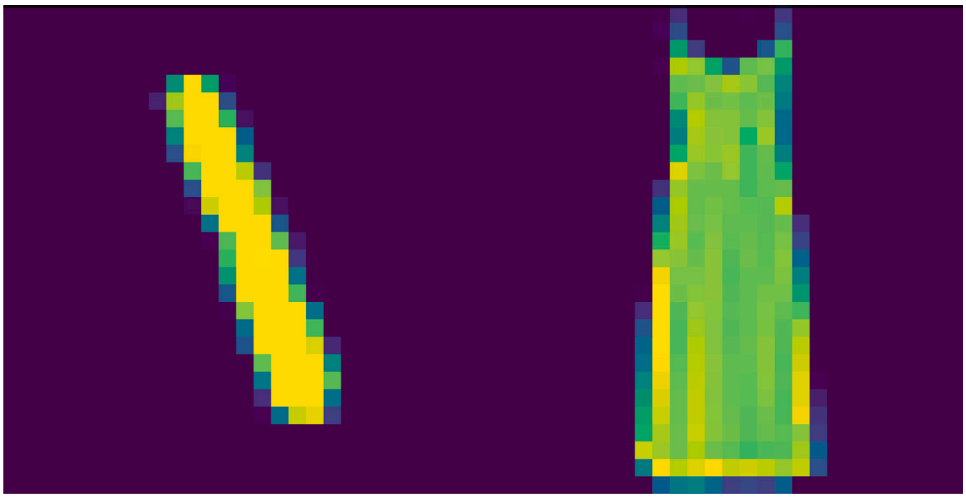


Fig. 8. MNIST image concatenated with FMNIST image.

size remains small, this overhead remains manageable and represents a controlled trade-off between computation and improved generalization.

9. Conclusion

Limited data remains a fundamental challenge for deep neural networks, often leading to poor generalization when models are trained from scratch. While data augmentation and transfer learning provide partial remedies, their effectiveness may be constrained by issues such as negative transfer, domain mismatch, and reliance on large external datasets. To address these challenges, this work introduced a simple yet effective Contrast-Based Learning (CBL) framework that promotes relational feature learning by spatially pairing training samples. By exposing the network to pairwise contrasts, CBL enhances discriminative representation learning without requiring architectural modifications or loss-function redesign.

Extensive experiments across diverse datasets, including MNIST, Fashion-MNIST, CIFAR-10, COVID-19 chest X-rays, pollen-bearing honeybee images, and the Iris dataset, demonstrate the effectiveness and generality of the proposed approach. Comparative evaluations against standard training, transfer learning, and augmentation-based baselines (RandAugment, Mixup, and CEGAN) show consistent performance improvements in limited-data regimes. Interpretability analyses using qualitative and quantitative Grad-CAM evaluation further reveal that CBL encourages contrast-driven and semantically meaningful feature utilization.

Ablation studies on control-set construction and the parameter L highlight the role of majority voting stability and reveal that moderate control-set sizes provide a favorable trade-off between accuracy and inference complexity. Additionally, ranking-based statistical validation using the Friedman test confirms the robustness and consistent superiority of CBL across datasets. Although the proposed method introduces additional inference overhead due to concatenation and voting, this cost remains manageable in small-data scenarios and represents a controlled trade-off for improved generalization.

9.1. Future work

Several promising directions emerge from this study. First, extending CBL to ultra high-resolution imaging modalities, such as digital pathology, through memory-aware or feature-level concatenation strategies represents an important avenue. Second, integrating CBL with few-shot and meta-learning frameworks may further improve generalization in extreme low-data regimes. Third, exploring adaptive control-set selection mechanisms and dynamic tuning of the parameter L could enhance inference efficiency. Fourth, combining CBL with privacy-preserving and federated learning paradigms offers potential for deployment in decentralized and data-sensitive environments. Finally, investigating theoretical foundations of relational contrast learning and its interaction with representation geometry remains an open research direction.

10. Control set selection and ablation on L

The test-time majority voting mechanism in CBL relies on a subset of training images, denoted as D^{sub}_{train}, referred to as the control set. In this work, D^{sub}_{train} is constructed using class-stratified random sampling, where L images are selected uniformly at random from each class. Stratification ensures balanced representation across all classes and prevents bias toward dominant categories.

Since sampling is performed independently for each class, the size of the control set becomes $|D^{sub}_{train}| = p \times L$, where p is the number of classes. For reproducibility, experiments are conducted with fixed random seeds, and results are averaged across multiple runs.

10.0.1. Effect of L on classification performance

To analyze the sensitivity of the majority voting mechanism, an ablation study was conducted on the 10-class sMNIST dataset by varying the number of control samples per class (L). Since the parameter L determines the number of pairwise comparisons performed during inference, increasing L improves voting stability but linearly increases computational cost.

All experiments were performed using identical model architecture, hyperparameters, and data splits, with only the value of L varied. The results are summarized in Table 11.

The findings indicate that classification accuracy improves as L increases from small values ($L = 1$) to moderate values ($L = 3$ and $L = 5$), after which performance gains begin to saturate. This behavior suggests that a limited number of control samples per class is sufficient to capture discriminative relational contrast information and stabilize majority voting decisions.

Furthermore, the effect of control set construction was evaluated by comparing stratified random sampling with non-stratified sampling. Stratified selection consistently produced more stable and higher accuracy results by ensuring balanced class representation during voting.

Overall, the ablation study demonstrates that:

- Moderate values of L provide the best trade-off between inference cost and predictive accuracy.
- Stratified control set selection improves robustness, particularly in low-shot settings.

Table 11
Effect of L on test accuracy on 10-class sMNIST.

L	Accuracy (%)
1	87.2
3	92.8
5	94.1
7	94.3

- Increasing L beyond moderate values yields diminishing returns while increasing computational overhead.

Ethics approval

This article contains no studies with human participants or animals performed by any of the authors.

Funding

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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